Alberta Public Health Disease Management Guidelines

Malaria

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Communicable Disease Control Branch

Public Health Division

Alberta Health

Malaria | Alberta Health, Government of Alberta

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Revisions

Revision Date	Document Section	Description of Revision
March 2025	General	• Entire guideline updated to align with current information.
	Case Definition	• Revised case definition to include “**and/or”** detection by PCR
	Clinical Assessment and Epidemiology	• Expanded information on Reservoir, Incubation period and Incidence

Case Definition

Confirmed Case

Laboratory confirmation of infection with or without clinical illness:[(A)]

- Demonstration of *Plasmodium* sp. in a blood smear/film (thick and thin)

AND/OR

- Detection of *Plasmodium* sp. nucleic acid (e.g., PCR) in an appropriate clinical specimen[(B)]

Probable Case

Laboratory evidence of infection with or without clinical illness:[(A)]

- Detection of *Plasmodium* sp. antigen[(C)] in an appropriate clinical specimen

(A) Clinical illness: Symptoms vary and include fever, headache, back pain, chills, sweats, myalgia, nausea, vomiting, diarrhea, and cough. See Clinical Presentation for more information.

(B) Refer to the Public Health Laboratory (ProvLab) Guide to Services for current specimen collection requirements.

(C) Including Rapid Antigen Detection Testing (RDT).

Reporting Requirements

Physicians, Health Practitioners and Others

- Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases by mail, fax, or electronic transfer within 48 hours (two business days).

Laboratories

- All laboratories shall report all positive laboratory results by mail, fax, or electronic transfer within 48 hours (two business days) to the:
  - Zone MOH (or designate), and
  - Chief Medical Officer of Health (CMOH) (or designate).

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- The Zone MOH (or designate) where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country cases, the following information should be forwarded to the CMOH (or designate) by phone, fax, or electronic transfer within 48 hours (two business days):
  - name,
  - date of birth,
  - out-of-province health care number,
  - out-of-province address and phone number,
  - positive laboratory report (if applicable), and
  - other relevant clinical/epidemiological information.

Clinical Assessment and Epidemiology

Etiology

Malaria is caused by protozoan parasites of the genus *Plasmodium*.[1,2] Five species of the disease are known to infect humans: *P. falciparum, P. vivax, P. ovale, P. malariae* and *P. knowlesi.*.[3–5]

Clinical Presentation

Malaria symptoms can range from uncomplicated to severe disease.[6,7] Both forms of infection begin with non-specific symptoms, including fever, chills, headache, muscle aches and weakness, abdominal pain, diarrhea, cough, and nausea/vomiting.[3,6,8,9] Individuals with established infection will typically experience a cycle of fever-free intermissions with symptoms of chills, fever and sweating recurring at various intervals depending on the species.[1,3] Malaria in young children, infants or those with congenital malaria may present with irritability, lethargy, refusal to eat, and vomiting.[3,4]

Individuals who have grown up in endemic areas and acquired partial immunity or individuals who are non-immune and who have taken prophylactic antimalarial medications, may have an atypical clinical presentation (such as acute renal failure and acute respiratory distress).[3,10] Pregnant individuals, especially primigravid persons in endemic areas, infected with the malaria parasite are at an increased risk for spontaneous abortions or stillbirth.[4]

P. vivax, P. ovale, P. malariae, and *P. knowlesi* are not usually considered to be life threatening.[3] *P. vivax* and *P. ovale* can cause relapses at irregular intervals, in rare cases up to 4 years.[3]

Severe malaria is life-threatening and most often caused by *P. falciparum* (and, rarely, other species) and may manifest as: cerebral malaria (impaired consciousness and/or seizures), severe anemia, circulatory shock, acute respiratory distress syndrome (ARDS/pulmonary edema), abnormalities in blood coagulation/significant bleeding, jaundice, hemoglobin in urine ("blackwater fever") acute kidney injury, hypoglycemia, hyper-parasitemia, and metabolic acidosis.[6]

The case fatality rate of uncomplicated *P. falciparum* malaria in non-immune individuals and children is 0.3%, however case fatality increases to 15 to 20% if complications develop.[3] Untreated *P. falciparum* can quickly progress in severity and result in death, with 80% of fatal malaria cases due to cerebral malaria.[3,11,12]

Diagnosis

Laboratory confirmation is made by direct microscopic examination of intracellular parasites on stained blood films (e.g., smear) and/or by PCR testing.[3] Due to the variable performance of rapid antigen detection tests (RDT) and low incidence of malaria in Alberta, RDT results need to be confirmed by PCR and smear.[D]

Treatment

Prompt treatment is essential, specifically for *P. falciparum* malaria, as severe complications can rapidly appear.[1,3,4,13] The choice of treatment is based on the patient's age, the infecting species, the severity of disease, and possible drug resistance.[1–3]

[D] Personal communication with Dr. Hong Yuan Zhou and Dr. King Kowalewska-Grochowska March 28, 2023, and April 8, 2023.

Reservoir

The only known reservoir of *P. falciparum, P. vivax, P. ovale* and *P.malariae* malarias is humans, although *P. malariae* occurs in African apes and probably some South American monkeys.[3] *P. knowlesi* is a true zoonosis and has been identified in some areas of Southeast Asia with long and pig-tailed macaque monkeys as the natural host.[3]

Transmission

Malaria is transmitted to humans via the bite of an infected female mosquito of the *Anopheles* genus.[1–4,14,15] Humans can develop sufficient viremia to infect feeding mosquitos which can result in further transmission to other humans during subsequent feedings.[3,14] Rarely, transmission of malaria can occur via the following:[1,3,14]

- Contaminated blood/blood products,
- Contaminated injection equipment or needlestick injury,
- Solid organ transplantation,[16] and
- In utero (congenital malaria)

Incubation Period

Varies depending on species with approximate ranges as follows:[3]

- *P. falciparum:* 6 to 14 days,
- *P. ovale:* 12 to 18 days,
- *P. vivax:* 12 to 18 days, may extend up to 6 to 12 months in temperate regions,
- *P. knowlesi:* 9 to 12 days,
- *P. malariae:* 18 to 40 days, parasites may persist for years in blood cells before being activated to cause symptoms.

The incubation period for infection by blood and blood products depends on the number of parasites infused and may take up to two months.[3]

Individuals who have grown up in endemic areas and acquired partial immunity or individuals who are non-immune and who have taken prophylactic antimalarial medications may have a prolonged incubation period.[3]

Period of Communicability

Malaria cases have the ability to infect other people via blood as long as the infective gametocytes continue to circulate.[3,14] The period of infectivity varies between parasite species.[3] If untreated or inadequately treated, cases can be infectious for decades (*P. malariae)* or up to five years (*P. Vivax.)*[3] *P. falciparum* cases can be infectious for up to one year, with potential to extend infectivity if reinfected in this timeframe.[3]

Host Susceptibility

Susceptibility is universal.[3] Individuals who are at higher risk for severe disease, especially with *P. falciparum* include:[1,3,4,14]

- Nonimmune people, especially young children (under 5 years of age),
- Pregnant individuals,
- Immunocompromised people.

In endemic areas, where infection rates are high:

- People may develop partial immunity after repeat infections.[3]

- Some genetic traits may alter susceptibility to malaria or the disease presentation.[3]
- Until they have acquired immunity, children remain highly vulnerable to infection, as do travellers to endemic areas.[3,4,17]

Incidence

Incidence of malaria is disproportionately high in the African regions, accounting for approximately 95% of all cases and 96% of deaths.[3,15] Children under 5 years are a significantly vulnerable group and account for 80% of all malaria related deaths in Africa.[3,15] For more information on worldwide incidence, refer to the PAHO/WHO website.

Malaria is nationally notifiable in Canada and annual incidence can be reviewed on the Government of Canada Notifiable Disease Charts.

Malaria first became notifiable in Alberta in 1985 and is not considered endemic in the province. Case counts for Alberta are available through the Interactive Health Data Application (IHDA).

Public Health Management

Key Investigation

- Confirm that the person meets case definition.
- Obtain a history of illness, including the date of onset of signs and symptoms.
- Determine where the case may have been exposed, taking into consideration the incubation period, reservoir, and mode of transmission, including a history of:
  - travel to and/or residence within the last six months in an area with endemic transmission or risk of malaria (See Incidence)
  - transfusion of blood or blood products, organ transplant recipient, or needlestick injury/use of contaminated injection equipment (within previous 60 days),[3]
  - previous malarial infection,
  - use of antimalaria chemoprophylaxis
- Determine if case is pregnant.

Management of a Case

- Prompt diagnosis, accurate identification of species, timely treatment and appropriate initial management is essential to prevent malaria-associated morbidity and mortality.[1,3,4,15,18]
- All cases should be advised on disease transmission and prevention.
- Consultation with an Infectious Disease specialist is strongly recommended.

Management of Contacts

- There is no direct follow-up of contacts; however, it may be prudent to identify others who may have been exposed to the same source as the case to find undiagnosed cases.
- Individuals exposed to the same source should be advised of the following:
  - signs, symptoms, and prevention,
  - to monitor for symptoms and if symptoms occur to see their healthcare provider for assessment.

Preventive Measures

- Provide the following recommendations to travellers going to malaria-endemic areas:
  - consult a health care provider or visit a travel health clinic before travel.
  - refer to the Public Health Agency of Canada (PHAC) Travel health Notices and Health advice for travellers.
  - educate travellers (especially those who are pregnant or with young children) going to malaria-endemic areas on the following:[2–4,15]
  - there is currently no vaccine available in Canada for prevention of malaria.[19]
  - use personal protective measures to reduce mosquito exposures.
    - information on use of mosquito/insect repellents is available from Health Canada and the Canadian Pediatric Society.
  - for additional guidance, refer to My Health Alberta Malaria and Prevention of Malaria.

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